Introduction

There are few things I want to understand in this analysis so we can understand more about the data-field job market in LinkedIn, by inspecting jobs' requirements, salaries and other relevant info that might help us paint a clear picture about the current market of data-field jobs in LinkedIn.

> **NOTE**: Data-field refers to tech jobs that work mainly with data which are:
>
> - Data analysts
> - Data engineers
> - Data scientests
> - Machine learning developers

Things we are analysing in this notebook:

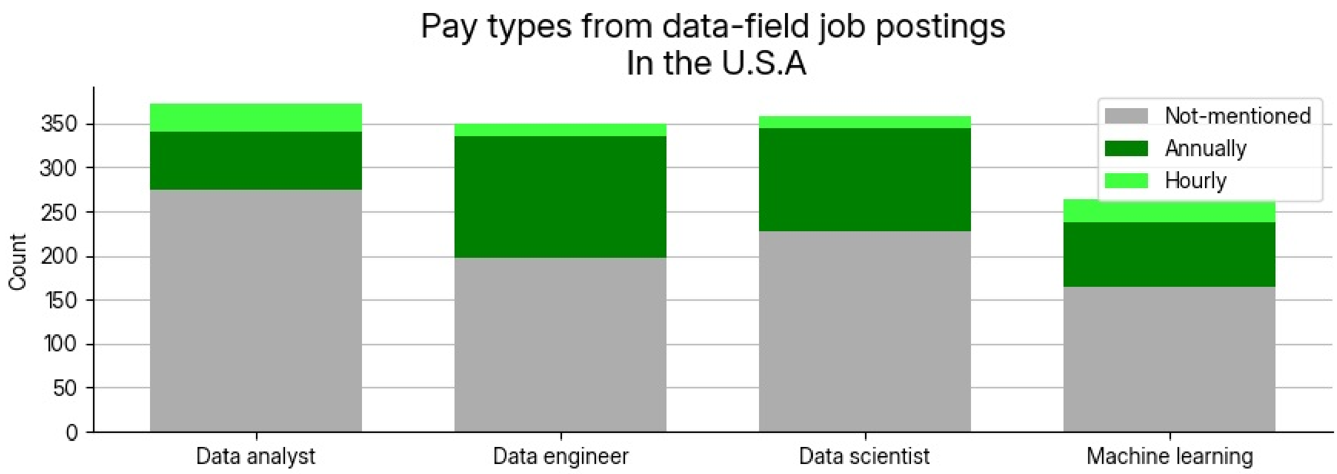
- Q1. Pay types
- Q2. Average salary
- Q3. Frequent locations
- Q4. Mentioned skills
- Q5. Common industries
- Q6. Required education
- Q7. Seniority level

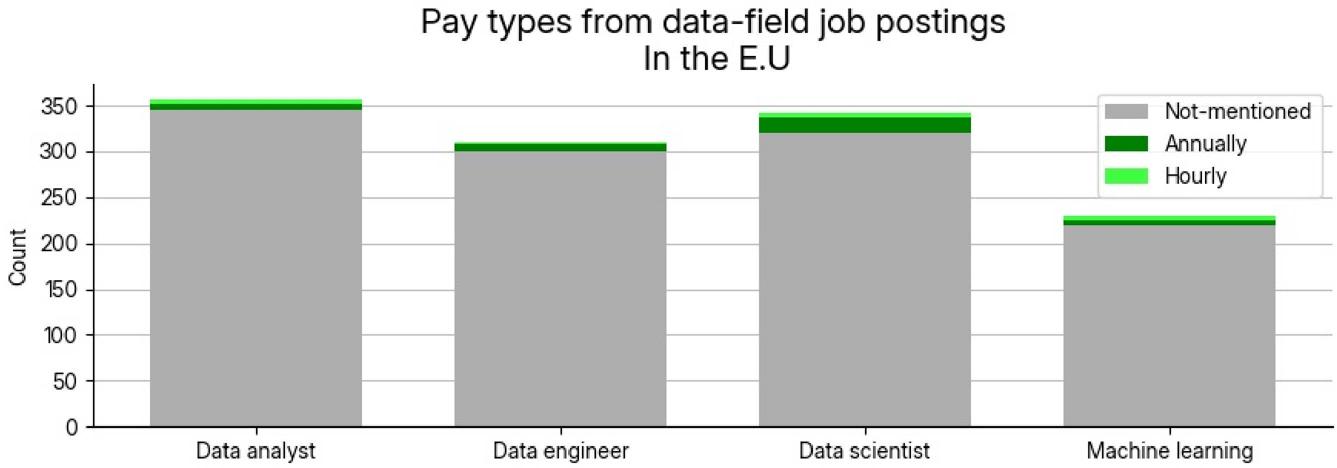
Setting up

Analysis

Q1. Pay types

Firstly we are gonna see how common each pay type in the description is.
pay type can have a one of the following values: annauly, hourly, or not-mentioned.
This will help us know how common it is for companies to post thier salary and how common is hourly payment in data-field jobs





Oh this wasn't expected, Most of the job postings don't include salary specially jobs postings in the E.U

*(This might also be because that the salary extractor works best with English)

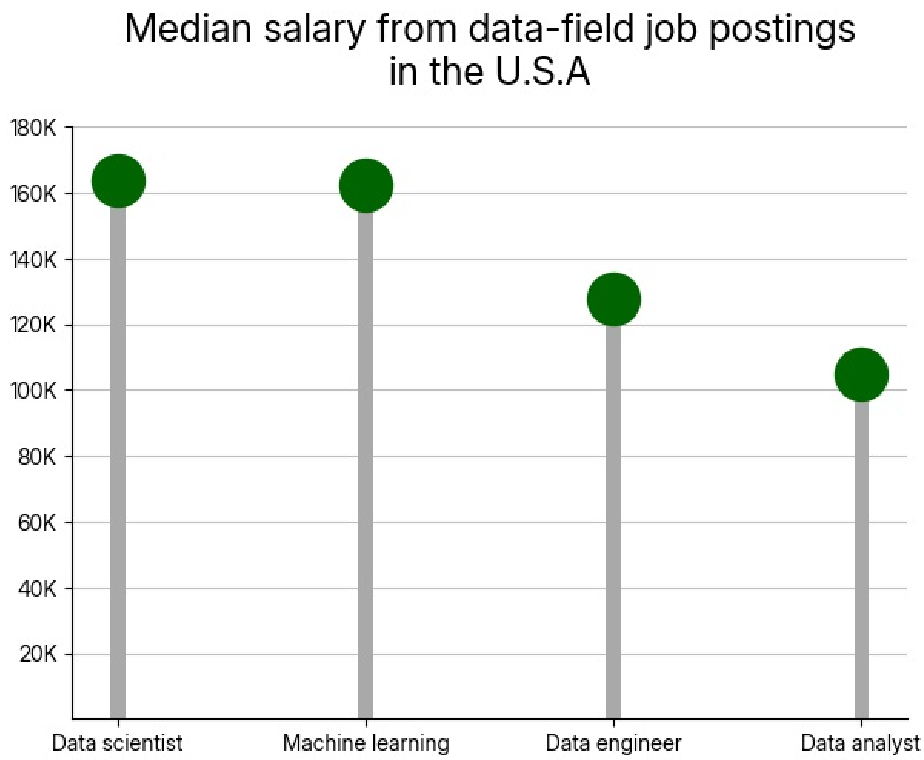
This also means that we are not going to be able to analyse average salary in E.U job postings.

We can also see that data engineers & scientists are paid annauly most of the time,
On the other hand we can see that data analysts and machine learning devs payment can be hourly more often.

Q2. Average salary

Let's now see the median of mentioned salaries for data-field jobs in the USA.

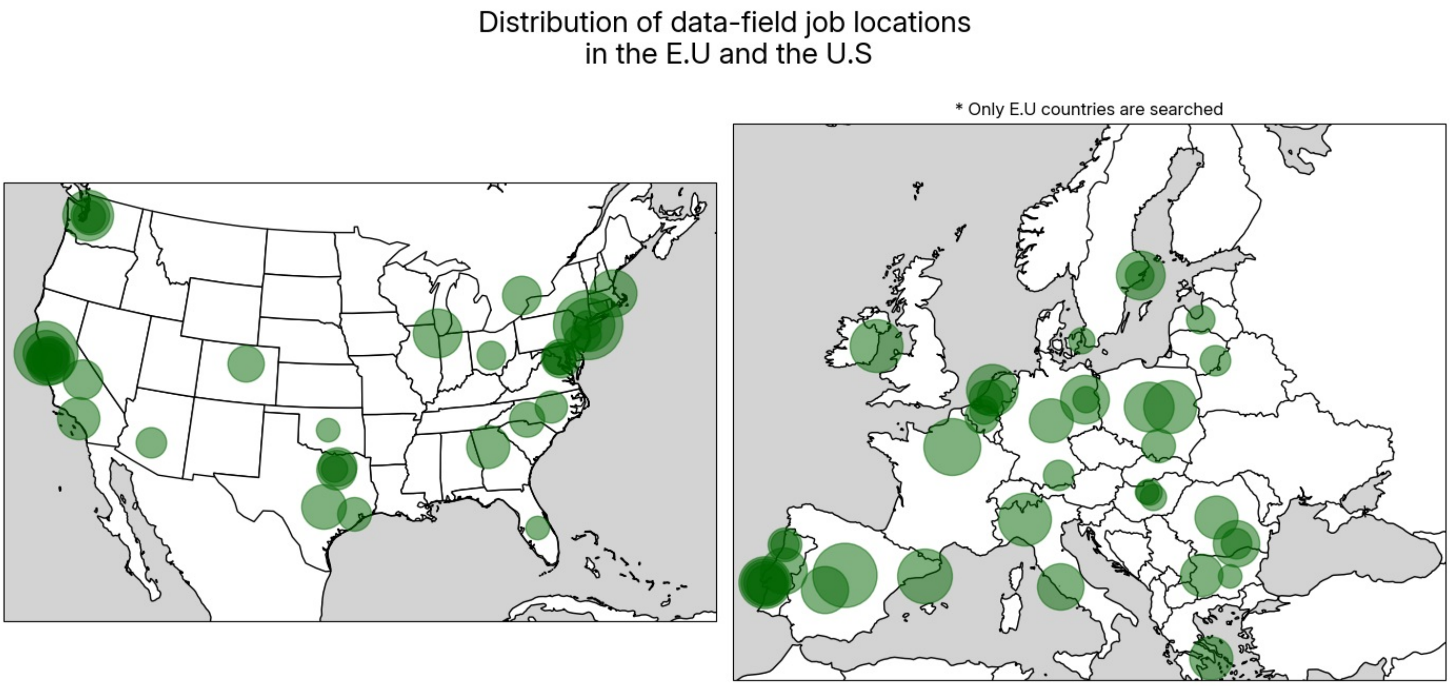
NOTE: As seen in the previous question Q1 most of the jobs doesn't mention salary at all and those who do isn't always annaul **SO** the result will only show the average salaries for jobs which mentions it in LinkedIn



We can see that all of those average salaries are really high compared to average salaries for the same jobs in the USA which makes senese because companies that offer high salaries are more likally to mention it in their job descriptions.

Q3. Frequent locations

Now let us look at where data-jobs are most needed.



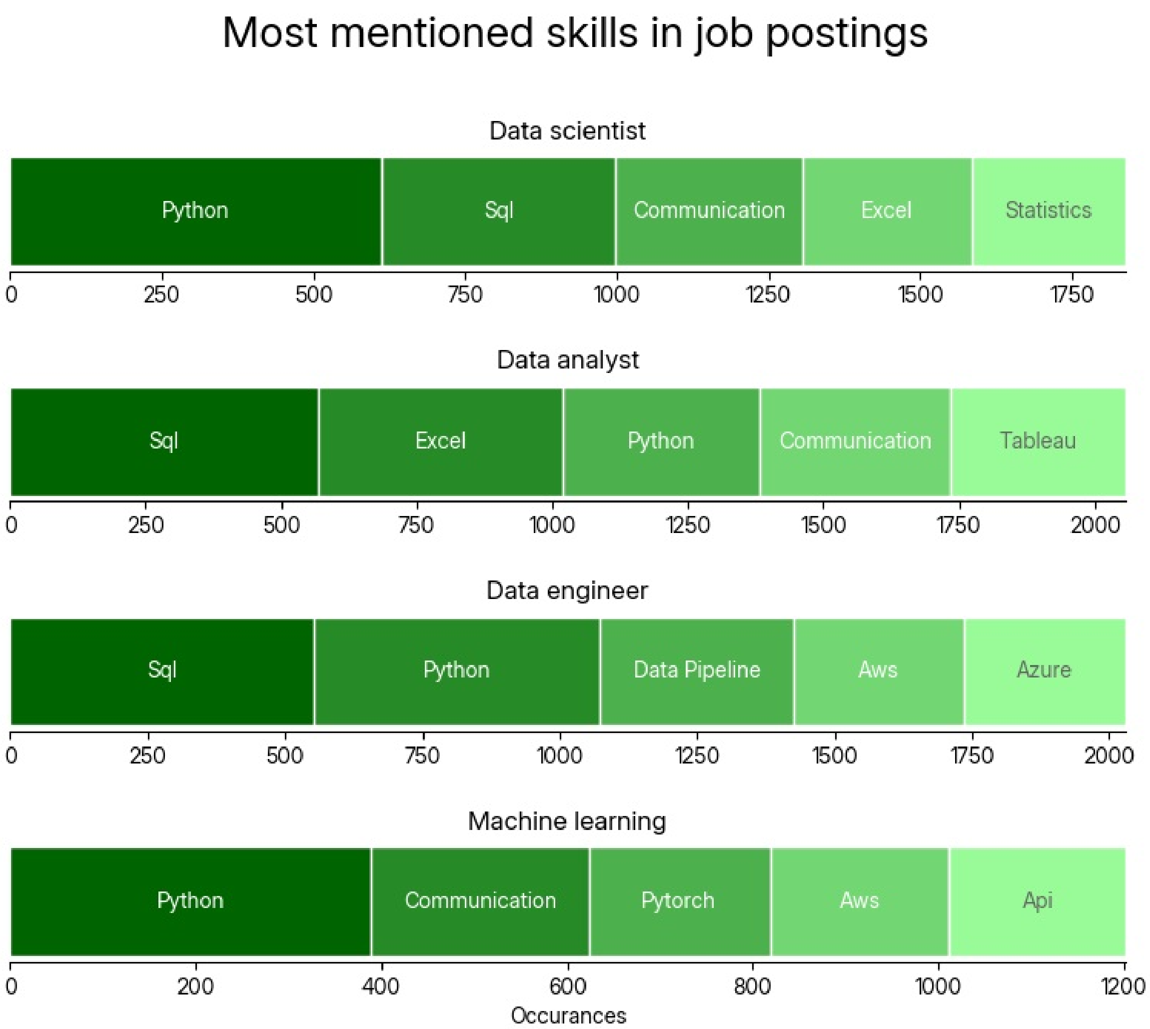
*Radius of the frequency circles is scaled logarithmically

As expected big cities and tech companies areas such as Silicon Valley, NY require data-jobs a lot.

What wasn't expected is that some eastren europe countries such as Poland and Romania also require data-jobs a lot.

Q4. Mentioned skills

What are the most mentioned skills in jobs descriptions for each data-field job title answering this question will show us the most important skills to have to apply to any of these data-field jobs.

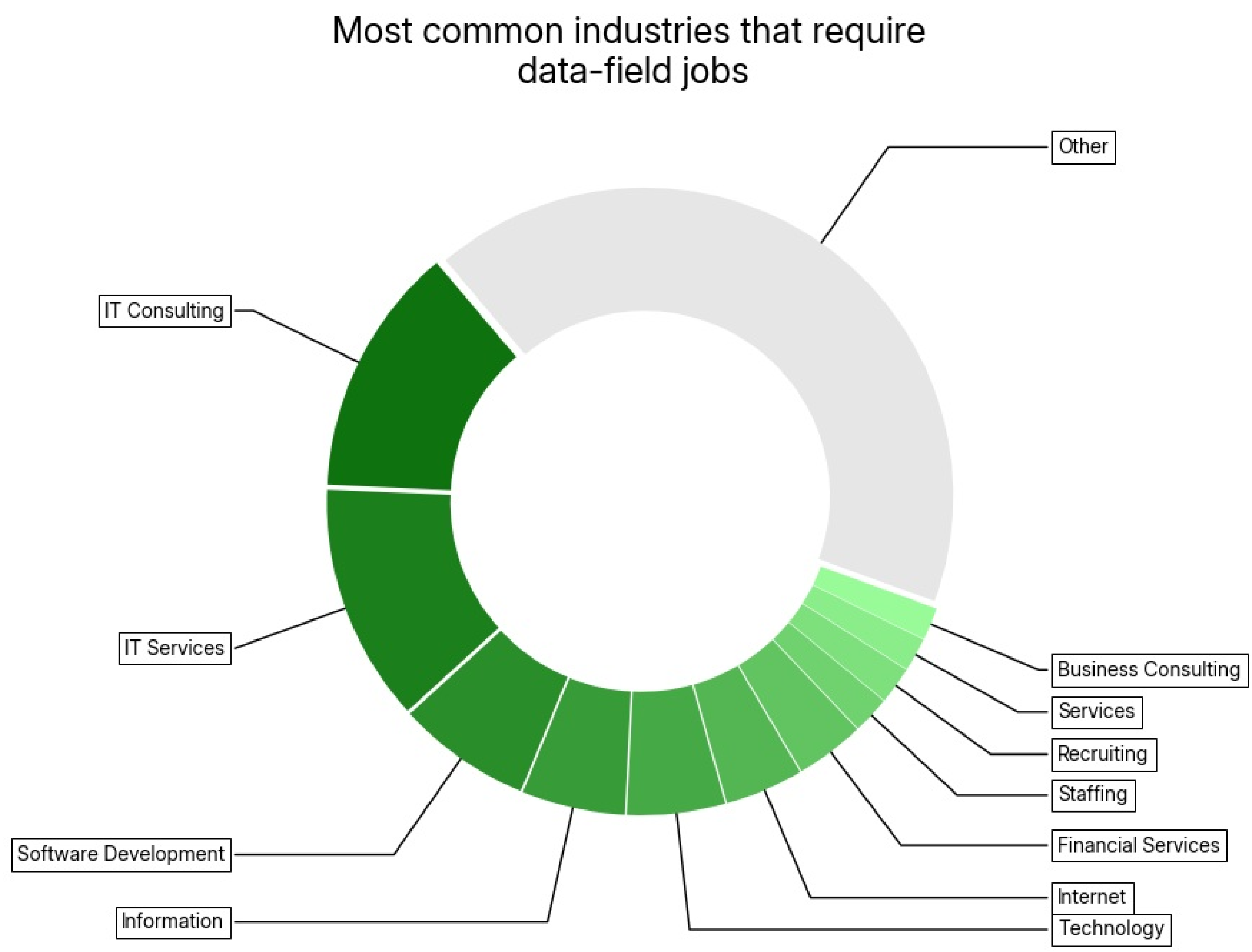


Sorry if the chart look kind of misleading but this is the most suitable chart type I found, anyway as I expected SQL and Python the most requested.

Also we can see that communication is frequently requested which makes sense also.

Q5. Common industries

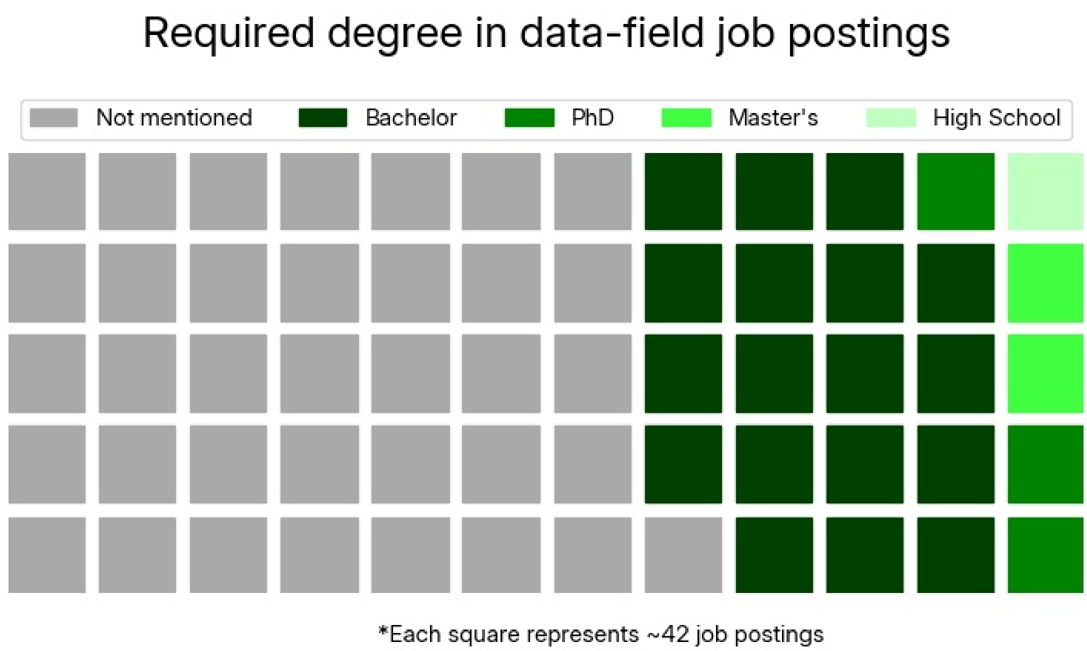
Looking at which industries employ data-field jobs the most will let us know what background knowledge can be valuable when applying to a data-field job.



Looks like most of these industries are tech related so it might be useful to have background I.T or software development, other industries that require data-field jobs are side jobs such as recruiting and financial stuff.

Q6. Required education

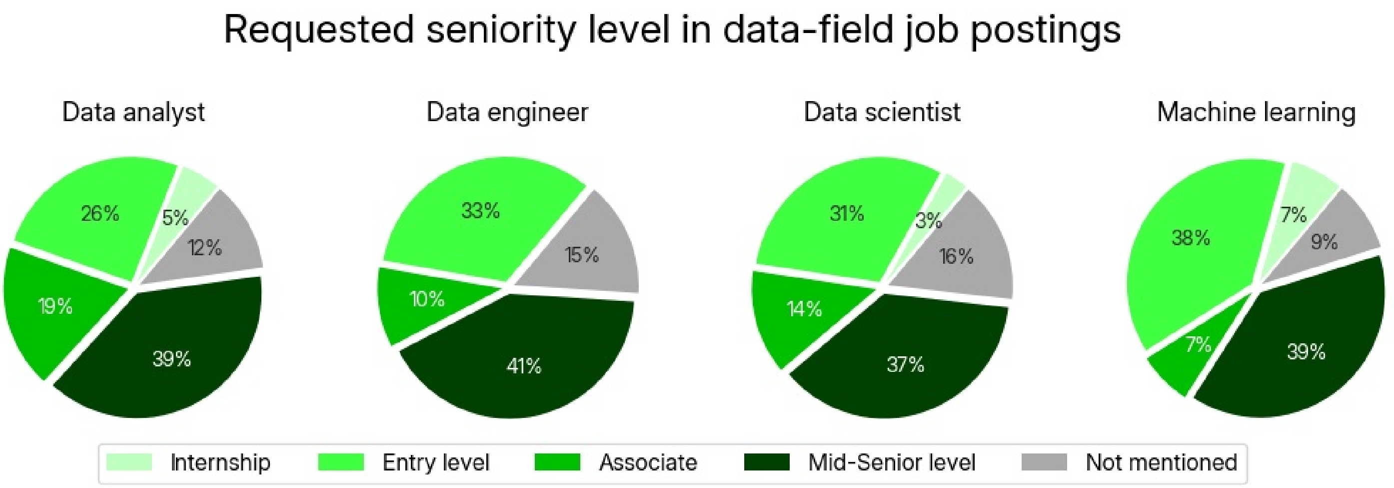
What's the most required degree that data-field jobs ask for, This is the question I am most curious to answer, It'll help us see how much do companies care for a candidate to have a degree



Wow that surprising that more than 50% of the jobs didn't even mention a degree in thier job description.

Q7. Seniority level

Finally let's look at what's the most requested seniority levels by data-field jobs
which will help us see what data-field job is the most open to newcomers.



We can see that Machine learning jobs are the most open to juniors and seniors in the same time, this might also be due it's the job with the least amount of N/A values in the seniority level.

But -surprisingly- data analyst jobs are the least open to Entry level though some of them offers internships.

Summary

Thanks 😊 for looking into my humble analysis for the data-field job market
hope you found it informative here's a summary of all of the findings:

- Q1. Pay Types:
  Most job postings do not mention salary, especially in Europe. When salaries are listed, data engineers and scientists are usually paid annually, while analyst and ML roles more often mention hourly rates.
- Q2. Average Salary (USA):
  The median salaries shown are high, likely because only companies offering competitive pay choose to include salaries in job descriptions—so the results are biased toward higher-paying roles.
- Q3. Frequent Locations:
  Major tech hubs like Silicon Valley and New York dominate demand for data roles, but unexpectedly high demand was also seen in Eastern European countries such as Poland and Romania.
- Q4. Most Mentioned Skills:
  SQL and Python are the top required skills across all data roles, with communication skills also appearing frequently.
- Q5. Common Industries:
  Most opportunities come from tech-related industries, followed by fields like finance and recruiting, suggesting that an IT or software background is especially valuable.
- Q6. Required Education:
  Over 50% of postings do not specify a degree requirement, indicating that many companies are flexible about formal education, relying instead on skills and experience.
- Q7. Seniority Levels:
  Machine learning roles show strong demand for both junior and senior levels. Data analyst roles, however, are less open to entry-level candidates, despite some internship availability.